

JT1078 Data Sending Protocols

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Version History

Version	Date	Modifier	Changes
1.0.0	2026-05-06	Zhuojie Cai	Initial version
1.0.1	2026-05-06	Zhuojie Cai	Added version difference notes (2013 vs 2019)
1.0.2	2026-06-11	Zhuojie Cai	Added functional descriptions for 0x9201/0x9202/0x1206/0x9207 messages

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1 Protocol Overview

JT/T 1078 is the communication protocol and data format standard between vehicle-mounted video terminals and video platforms in the road transport vehicle satellite positioning system. It adopts **TCP/UDP** communication. Real-time audio/video transmission uses **newly established links**, while signaling data reuses the established positioning information link of JT/T 808.

Protocol Inheritance Note: JT/T 1078 inherits all instructions from JT/T 808 except 0x8804 (Recording Start Command), and adds video-related instructions.

Message Direction Rules:

- Uplink messages (Terminal → Platform): Message IDs start with **0x1**, such as 0x1003, 0x1205
- Downlink messages (Platform → Terminal): Message IDs start with **0x9**, such as 0x9003, 0x9101

Message Classification:

- Signaling data messages: Reuse JT/T 808 positioning link
- Stream data messages: Newly established links for audio/video stream transmission

Version Differences (2013 vs 2019):

- Protocol version 2019 extends the terminal phone number from **BCD[6]** to **BCD[10]**, corresponding to the "Terminal Device SIM Card Number" field in RTP data: Protocol version 2013 uses **6 bytes** (BCD encoding), Protocol version 2019 uses **10 bytes** (BCD encoding)

Terminology

Term	Full Name/Meaning	Description
AAC	Advanced Audio Coding	Advanced Audio Coding
MPEG	Moving Pictures Experts Group	Moving Pictures Experts Group
RTP	Real-time Transport Protocol	Real-time Transport Protocol
TCP	Transmission Control Protocol	Transmission Control Protocol
UDP	User Datagram Protocol	User Datagram Protocol
URL	Uniform Resource Locator	Uniform Resource Locator
UTF-8	Unicode Transformation Format	Unicode Transformation Format
FTP	File Transfer Protocol	File Transfer Protocol
CBR	Constant Bit Rate	Constant Bit Rate
VBR	Variable Bit Rate	Variable Bit Rate
ABR	Average Bit Rate	Average Bit Rate
FPS	Frame per Second	Frames per Second
OSD	On-Screen Display	On-Screen Display

1.1 Real-time Audio/Video Transmission Channel Agreement

One real-time audio/video transmission channel can transmit one video stream or one audio stream, and can also transmit one video stream and one audio stream simultaneously. The real-time audio/video transmission channel agreements are as follows:

- When using TCP, each TCP connection can carry multiple audio/video channels. If there is no data transmission within the specified timeout period, both the terminal and the monitoring center can actively close the TCP connection used for audio/video data transmission
- When using UDP, each UDP port can carry multiple audio/video channels

1.2 Audio/Video Communication Message Classification

Audio/video data messages are divided into the following two categories:

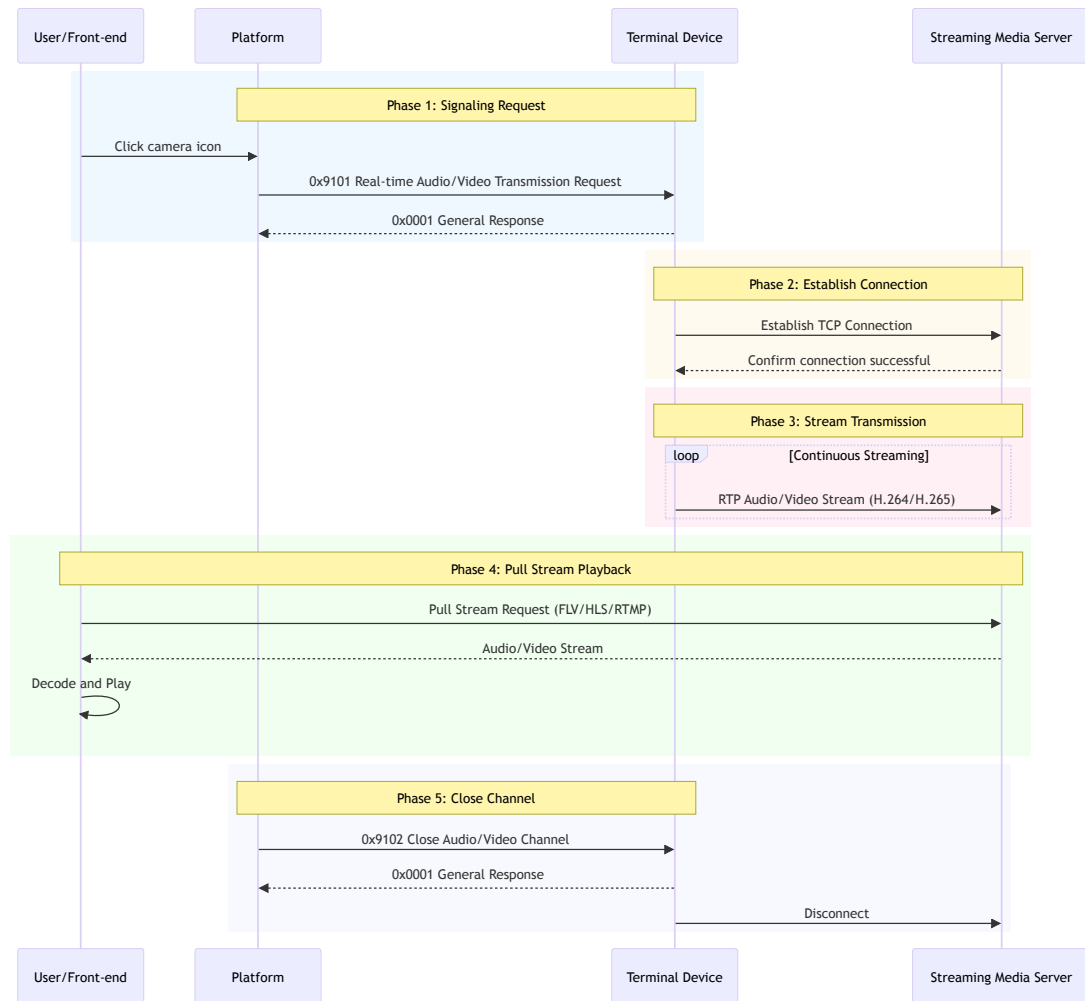
- **Signaling data messages:** The data format shall comply with the provisions of JT/T 808. On this basis, new protocol instructions and data formats are added. Message communication shall use the existing link between the vehicle-mounted video terminal and the enterprise video monitoring platform for transmitting positioning information, and shall not establish new links
- **Stream data messages:** Used for network real-time audio/video transmission, network video recording playback, voice intercom, audio monitoring, voice broadcasting, etc. Message communication shall establish new links, and shall not use the link for transmitting positioning information

1.3 Real-time Video Request Process

Real-time video transmission adopts a signaling and stream separation architecture, as follows:

Phase	Operation	Key Instruction	Description
Phase 1	Platform sends real-time audio/video transmission request	0x9101	Platform → Terminal, specifying streaming media server IP and port, logical channel number, stream type and other parameters

Phase 2	Terminal parses instruction and establishes connection	General Response	Terminal replies with general response, starts pushing audio/video stream to streaming media server
Phase 3	Terminal pushes audio/video stream to server	RTP Transmission	Device continuously sends encoded (H.264/H.265, etc.) audio/video data packets to streaming media server
Phase 4	Client pulls stream from server for playback	—	Front-end player obtains audio/video stream from streaming media server through FLV/HLS/RTMP protocols and decodes for playback
Phase 5	Close channel	0x9102	Platform sends close instruction, terminal stops pushing stream, releases resources



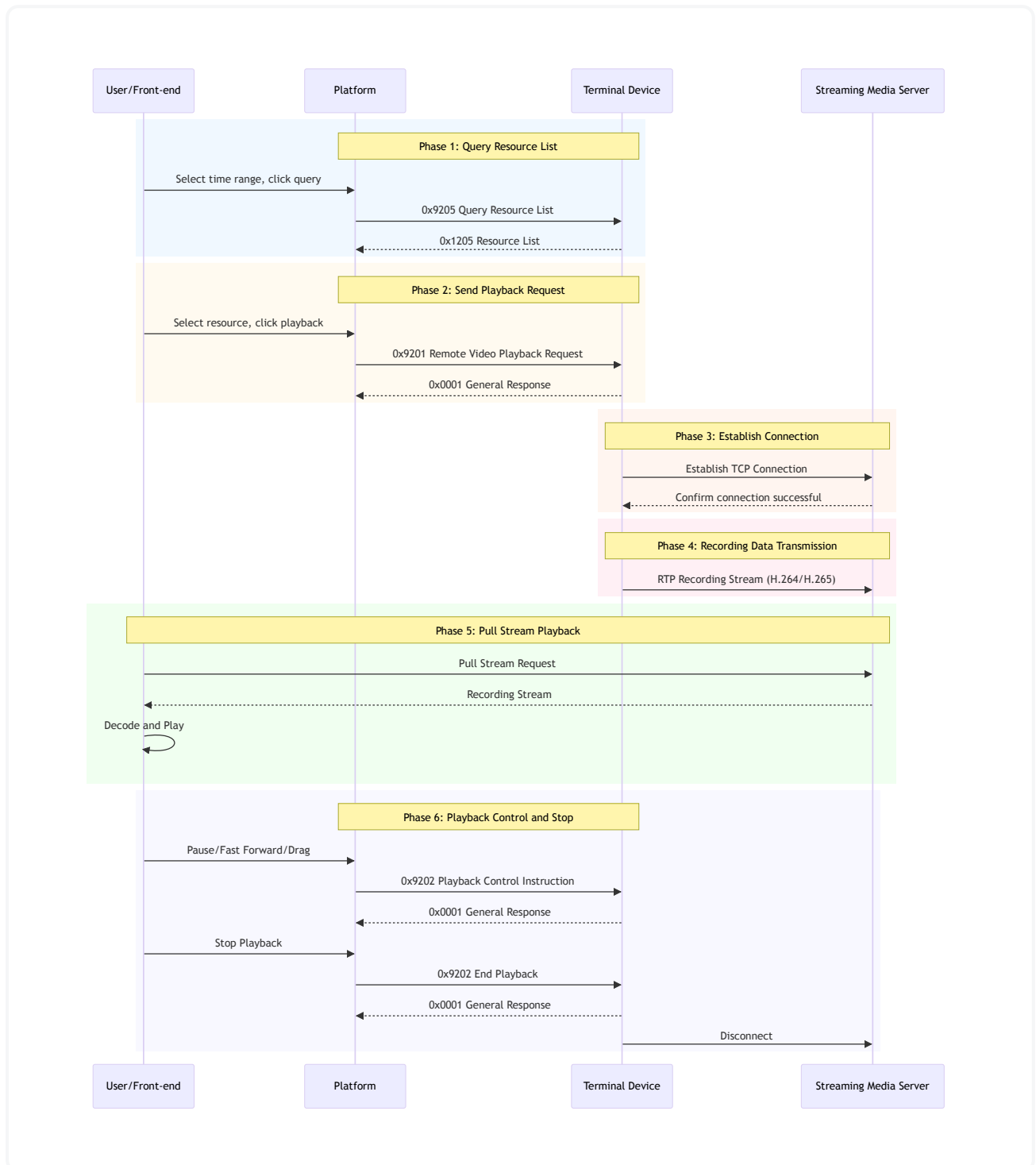
1.4 Historical Video Playback Process

Users select a time period for historical video playback, as follows:

Phase	Operation	Key Instruction	Description
Phase 1	Platform queries resource list	0x9205	Specify logical channel number, time range and other conditions to query available recordings

Phase 2	Terminal uploads resource list	0x1205	Terminal returns audio/video resource list matching the conditions
Phase 3	Platform sends remote video playback request	0x9201	After user selects resource, platform specifies streaming media server, logical channel number, time range, playback mode, etc.
Phase 4	Terminal parses request and establishes connection	General Response	Terminal replies with general response, reads recording from storage and establishes connection to streaming media server
Phase 5	Terminal transmits recording data to server	RTP Transmission	Recording data is continuously transmitted to streaming media server in H.264/H.265 format, etc.
Phase 6	Client pulls stream for playback	—	Front-end player obtains recording stream from streaming media server and decodes for playback

Phase 7	Playback control and stop	0x9202	Supports pause, fast forward, drag and other controls; platform sends stop instruction, terminal stops transmission and releases resources
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2 Protocol Basics

2.1 Data Types

Type	Description
BYTE	Unsigned single-byte integer (8 bits)
WORD	Unsigned double-byte integer (16 bits)
DWORD	Unsigned quadruple-byte integer (32 bits)
BCD[n]	8421 BCD encoding, n bytes
STRING	UTF-8 encoded string
BYTE[n]	n bytes of data

Transmission Rule: Big-endian mode, high byte first.

2.2 Message Structure

The signaling message structure of JT/T 1078 is consistent with JT/T 808:

Part	Description
Flag	Fixed 0x7E
Message Header	See JT/T 808 message header structure
Message Body	Customized by each message
Check Code	XOR checksum, 1 byte
Flag	Fixed 0x7E

Escape Rules: Escape processing must be strictly executed in sequence

1. **Step 1:** Escape **0x7D** in message content (check code, message header, message body) → **0x7D 0x01**
2. **Step 2:** Escape **0x7E** in the result of Step 1 → **0x7D 0x02**

Note: Process 0x7D first, then 0x7E. The order cannot be reversed.

Example: Original data `0x30 0x7E 0x08 0x7D 0x55` → Escaped `0x7E 0x30 0x7D 0x02 0x08 0x7D 0x01 0x55 0x7E`

3 Message Overview

3.1 Uplink Messages (Terminal → Platform)

Message ID	Name	Category	Remarks
0x1003	Terminal Uploads Audio/Video Attributes	Audio/Video Attributes	
0x1205	Terminal Uploads Audio/Video Resource List	Historical Audio/Video	
0x1206	File Upload Complete Notification	Historical Audio/Video	
0x120F	Recording Calendar Information Reply	Historical Audio/Video	
0x1215	Terminal Uploads Audio/Video Resource List + Event Point List	Historical Audio/Video	

3.2 Downlink Messages (Platform → Terminal)

Message ID	Name	Category	Remarks
0x9003	Query Terminal Audio/Video Attributes	Audio/Video Attributes	Message body is empty

0x9101	Real-time Audio/Video Transmission Request	Real-time Audio/Video	
0x9102	Audio/Video Real-time Transmission Control	Real-time Audio/Video	
0x9201	Platform Sends Remote Video Playback Request	Historical Audio/Video	
0x9202	Platform Sends Remote Video Playback Control	Historical Audio/Video	
0x9205	Query Resource List	Historical Audio/Video	
0x9206	File Upload Instruction	Historical Audio/Video	
0x9207	File Upload Control	Historical Audio/Video	
0x920F	Query Recording Calendar Information	Historical Audio/Video	
0x9215	Query Playback Resources and Event Point List	Historical Audio/Video	

4 Message Details

4.1 Parameter Settings

Audio/Video Parameter Settings

Terminal audio/video parameter settings adopt the 0x8103 message in JT/T 808, with the following audio/video parameter settings added:

Parameter ID	Data Type	Description
0x0075	—	Audio/Video Parameter Settings, see Table 2
0x0076	—	Audio/Video Channel List Settings, see Table 3
0x0077	—	Individual Video Channel Parameter Settings, see Table 5
0x0079	—	Special Alarm Recording Parameter Settings, see Table 7
0x007A	DWORD	Video-related alarm mask, corresponding to video alarm flag bit definitions; when the corresponding bit is 1, the corresponding type of alarm is masked
0x007B	—	Image Analysis Alarm Parameter Settings, see Table 8
0x007C	—	Terminal Sleep/Wake Mode Settings, see Table 9

Table 2 Audio/Video Parameter Definition and Description

Offset	Field	Data Type	Description
0	Real-time Stream Encoding Mode	BYTE	0: CBR 1: VBR 2: ABR 100~127: Custom

1	Real-time Stream Resolution	BYTE	0: QCIF 1: CIF 2: WCIF 3: D1 4: WD1 5: 720P 6: 1080P 100~127: Custom
2	Real-time Stream Key Frame Interval	WORD	Range (1~1000) frames
4	Real-time Stream Target Frame Rate	BYTE	Range (1~120) frames/s
5	Real-time Stream Target Bitrate	DWORD	Unit: kilobits per second (kbps)
9	Storage Stream Encoding Mode	BYTE	0: CBR 1: VBR 2: ABR 100~127: Custom
10	Storage Stream Resolution	BYTE	0: QCIF 1: CIF 2: WCIF 3: D1 4: WD1 5: 720P 6: 1080P 100~127: Custom
11	Storage Stream Key Frame Interval	WORD	Range (1~1000) frames
13	Storage Stream Target Frame Rate	BYTE	Range (1~120) frames/s
14	Storage Stream Target Bitrate	DWORD	Unit: kilobits per second (kbps)

18	OSD Subtitle Overlay Settings	WORD	Bit settings: 0 means no overlay, 1 means overlay bit0: Date and time bit1: License plate number bit2: Logical channel number bit3: Longitude and latitude bit4: Driving record speed bit5: Satellite positioning speed bit6: Continuous driving time bit7 ~ 10: Reserved bit11 ~ 15: Custom
20	Audio Output Enabled	BYTE	0: Disabled 1: Enabled

Table 3 Audio/Video Channel List

Offset	Field	Data Type	Description
0	Total Audio/Video Channels	BYTE	Represented by 1
1	Total Audio Channels	BYTE	Represented by m
2	Total Video Channels	BYTE	Represented by n
3	Audio/Video Channel Mapping Table	BYTE[4×(1+m+n)]	See Table 4

Table 4 Audio/Video Channel Mapping Table

Offset	Field	Data Type	Description
0	Physical Channel Number	BYTE	Starts from 1
1	Logical Channel Number	BYTE	See Appendix B
2	Channel Type	BYTE	0: Audio/Video 1: Audio 2: Video

Table 5 Individual Channel Video Parameter Definition and Description

Offset	Field	Data Type	Description
0	Number of Channels Requiring Individual Video Parameter Settings	BYTE	Represented by n
1	Individual Channel Video Parameter Settings List	BYTE[21×n]	See Table 6

Table 6 Individual Channel Video Parameter Settings

Offset	Field	Data Type	Description
0	Logical Channel Number	BYTE	See Appendix B
1	Real-time Stream Encoding Mode	BYTE	0: CBR 1: VBR 2: ABR 100~127: Custom

2	Real-time Stream Resolution	BYTE	0: QCIF 1: CIF 2: WCIF 3: D1 4: WD1 5: 720P 6: 1080P 100~127: Custom
3	Real-time Stream Key Frame Interval	WORD	Range (1~1000) frames
5	Real-time Stream Target Frame Rate	BYTE	Range (1~120) frames/s
6	Real-time Stream Target Bitrate	DWORD	Unit: kilobits per second (kbps)
10	Storage Stream Encoding Mode	BYTE	0: CBR 1: VBR 2: ABR 100~127: Custom
11	Storage Stream Resolution	BYTE	0: QCIF 1: CIF 2: WCIF 3: D1 4: WD1 5: 720P 6: 1080P 100~127: Custom
12	Storage Stream Key Frame Interval	WORD	Range (1~1000) frames
14	Storage Stream Target Frame Rate	BYTE	Range (1~120) frames/s
15	Storage Stream Target Bitrate	DWORD	Unit: kilobits per second (kbps)

19	OSD Overlay Settings	WORD	Bit settings bit0: Date and time bit1: License plate number bit2: Logical channel number bit3: Longitude and latitude bit4: Driving record speed bit5: Satellite positioning speed bit6: Continuous driving time bit7 ~ 10: Reserved bit11 ~ 15: Custom
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Table 7 Special Alarm Recording Parameter Definition and Description

Offset	Field	Data Type	Description
0	Special Alarm Recording Storage Threshold	BYTE	Special alarm recording occupies the percentage of main storage threshold, value range 1~99, default is 20
1	Special Alarm Recording Duration	BYTE	Maximum duration of special alarm recording, unit: minutes (min), default is 5

2	Special Alarm Marking Start Time	BYTE	Recording time marked before special alarm occurs, unit: minutes (min), default is 1
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Table 8 Image Analysis Alarm Parameter Definition and Description

Offset	Field	Data Type	Description
0	Vehicle Rated Passenger Capacity	BYTE	Rated passenger capacity of passenger vehicle; alarm is generated when video analysis result exceeds this value
1	Fatigue Level Threshold	BYTE	Video analysis fatigue driving alarm threshold; alarm is generated when exceeded

Table 9 Terminal Sleep/Wake Mode Settings Data Format

Offset	Field	Data Type	Description
0	Sleep/Wake Mode	BYTE	Bit settings bit0: Conditional wake bit1: Timed wake bit2: Manual wake

1	Wake Condition Type	BYTE	Valid when bit0 in Sleep/Wake Mode is 1 bit0: Emergency alarm bit1: Collision/rollover alarm bit2: Vehicle door open
2	Timed Wake Day Settings	BYTE	Bit settings bit0: Monday bit1: Tuesday bit2: Wednesday bit3: Thursday bit4: Friday bit5: Saturday bit6: Sunday
3	Daily Timed Wake Parameter List	BYTE[17]	See Table 10; time periods shall not overlap

Table 10 Daily Wake Parameter Definition

Offset	Field	Data Type	Description
0	Timed Wake Enable Flag	BYTE	Bit settings bit0: Time period 1 wake time enabled bit1: Time period 2 wake time enabled bit2: Time period 3 wake time enabled bit3: Time period 4 wake time enabled
1	Time Period 1 Wake Time	BCD[2]	HHMM, value range 00:00~23:59

3	Time Period 1 Close Time	BCD[2]	HHMM, value range 00:00~23:59
5	Time Period 2 Wake Time	BCD[2]	HHMM, value range 00:00~23:59
7	Time Period 2 Close Time	BCD[2]	HHMM, value range 00:00~23:59
9	Time Period 3 Wake Time	BCD[2]	HHMM, value range 00:00~23:59
11	Time Period 3 Close Time	BCD[2]	HHMM, value range 00:00~23:59
13	Time Period 4 Wake Time	BCD[2]	HHMM, value range 00:00~23:59
15	Time Period 4 Close Time	BCD[2]	HHMM, value range 00:00~23:59

4.2 Audio/Video Attributes

0x9003 Query Terminal Audio/Video Attributes (Downlink)

Message body is empty.

Example: 7E 90 03 00 00 12 34 56 78 90 12 00 01 18 7E

0x1003 Terminal Uploads Audio/Video Attributes (Uplink)

Table 11 Terminal Uploads Audio/Video Attributes Data Format

Offset	Field	Data Type	Description
0	Input Audio Encoding Method	BYTE	See Table 12
1	Input Audio Channel Count	BYTE	

2	Input Audio Sampling Rate	BYTE	0: 8 kHz 1: 22.05 kHz 2: 44.1 kHz 3: 48 kHz
3	Input Audio Sampling Bit Depth	BYTE	0: 8-bit 1: 16-bit 2: 32-bit
4	Audio Frame Length	WORD	Range 1~4,294,967,295
6	Audio Output Supported	BYTE	0: Not supported 1: Supported
7	Video Encoding Method	BYTE	See Table 12
8	Maximum Audio Physical Channel Count Supported by Terminal	BYTE	
9	Maximum Video Physical Channel Count Supported by Terminal	BYTE	

Example: 7E 10 03 00 0A 12 34 56 78 90 12 00 01 13 01 00 01 00 64 01 62 10 20 B6 7E

Table 12 Audio/Video Encoding Type Definition Table

Code	Name	Remarks
0	Reserved	
1	G.721	Audio
2	G.722	Audio
3	G.723	Audio
4	G.728	Audio

5	G.729	Audio
6	G.711A	Audio
7	G.711U	Audio
8	G.726	Audio
9	G.729A	Audio
10	DVI4_3	Audio
11	DVI4_4	Audio
12	DVI4_8K	Audio
13	DVI4_16K	Audio
14	LPC	Audio
15	S16BE_STEREO	Audio
16	S16BE_MONO	Audio
17	MPEGUDIO	Audio
18	LPCM	Audio
19	AAC	Audio
20	WMA9STD	Audio
21	HEAAC	Audio
22	PCM Voice	Audio
23	PCM AUDIO	Audio
24	AACLC	Audio
25	MP3	Audio
26	ADPCMIA	Audio
27	MP4AUDIO	Audio
28	AMR	Audio
29~90	Reserved	

91	Passthrough	System
92~97	Reserved	Video
98	H.264	Video
99	H.265	Video
100	AVS	Video
101	SVAC	Video
102~110	Reserved	
111~127	Custom	

4.3 Video Alarms

Video alarm reporting adopts the method of reporting simultaneously with position information, as supplementary information for 0x0200 position information reporting, extending the JT/T 808 supplementary information definition table.

Table 13 Supplementary Information Definition Table Extension

Supplementary Info ID	Supplementary Info Length	Description
0x14	4	Video-related alarm, DWORD, bit settings, flag bit definitions see Table 14
0x15	4	Video signal loss alarm status, DWORD, bit settings bit0 ~ bit31 represent logical channels 1~32 respectively When the corresponding bit is 1, it indicates that video signal loss occurred on that logical channel

0x16	4	Video signal occlusion alarm status, DWORD, bit settings bit0 ~ bit31 represent logical channels 1~32 respectively When the corresponding bit is 1, it indicates that video signal occlusion occurred on that logical channel
0x17	2	Storage device fault alarm status, WORD, bit settings bit0 ~ bit11 represent main storage devices 1~12 respectively bit12 ~ bit15 represent disaster backup storage devices 1~4 respectively When the corresponding bit is 1, it indicates that the storage device has a fault
0x18	2	Abnormal driving behavior alarm detailed description, WORD, definitions see Table 15

Table 14 Video Alarm Flag Bit Definition

Bit	Definition	Processing Description
0	Video signal loss alarm	Flag is maintained until alarm condition is cleared
1	Video signal occlusion alarm	Flag is maintained until alarm condition is cleared
2	Storage unit fault alarm	Flag is maintained until alarm condition is cleared

3	Other video equipment fault alarm	Flag is maintained until alarm condition is cleared
4	Passenger vehicle overcapacity alarm	Flag is maintained until alarm condition is cleared
5	Abnormal driving behavior alarm	Flag is maintained until alarm condition is cleared
6	Special alarm recording reaching storage threshold alarm	Cleared after receiving response
7~31	Reserved	

Table 15 Abnormal Driving Behavior Flag Bit Definition

Offset	Field	Data Type	Description
0	Abnormal driving behavior type	WORD	Bit settings bit0: Fatigue bit1: Phone call bit2: Smoking bit3~bit10: Reserved bit11~bit15: Custom
2	Fatigue level	BYTE	Fatigue level is represented by 0~100; higher values indicate more severe fatigue

4.4 Real-time Audio/Video Transmission

0x9101 Real-time Audio/Video Transmission Request (Downlink)

The platform requests real-time audio/video transmission from the terminal device, including real-time video transmission, actively initiating two-way voice intercom, one-way monitoring, broadcasting voice to all terminals, and specific passthrough.

Table 17 Real-time Audio/Video Transmission Request Data Format

Offset	Field	Data Type	Description
0	Server IP Address Length	BYTE	Length n
1	Server IP Address	STRING	Real-time video server IP address
1+n	Server Video Channel Listening Port (TCP)	WORD	Real-time video server TCP port number
3+n	Server Video Channel Listening Port (UDP)	WORD	Real-time video server UDP port number
5+n	Logical Channel Number	BYTE	See Appendix B
6+n	Data Type	BYTE	0: Audio/Video 1: Video 2: Two-way intercom 3: Monitoring 4: Center broadcast 5: Passthrough
7+n	Stream Type	BYTE	0: Main stream 1: Sub-stream

Example: 7E 91 01 00 15 12 34 56 78 90 12 00 01 0D 31 39 32 2E 31 36 38 2E 31 2E 31 30 30 1F 40 1F 41 01 00 01 29 7E

When the platform receives a special alarm from the video terminal, it shall proactively send this instruction to start real-time audio/video transmission without waiting for manual confirmation.

0x9102 Audio/Video Real-time Transmission Control (Downlink)

The platform sends audio/video real-time transmission control instructions, used for stream switching, pausing stream transmission, closing audio/video transmission channels, etc.

Table 18 Audio/Video Real-time Transmission Control Data Format

Offset	Field	Data Type	Description
0	Logical Channel Number	BYTE	See Appendix B
1	Control Instruction	BYTE	0: Close audio/video transmission instruction 1: Switch stream (add pause and resume) 2: Pause all stream sending on this channel 3: Resume stream sending before pause 4: Close two-way intercom
2	Close Audio/Video Type	BYTE	0: Close all audio/video data related to this channel 1: Only close audio related to this channel, keep video related to this channel 2: Only close video related to this channel, keep audio related to this channel
3	Switch Stream Type	BYTE	0: Main stream 1: Sub-stream

Example:7E 91 02 00 04 12 34 56 78 90 12 00 01 01 00 00 00 1D 7E

4.5 Historical Audio/Video

0x9205 Query Resource List (Downlink)

The platform queries video file lists from the terminal based on combined conditions such as audio/video type, channel number, alarm type, and start/end time.

Table 21 Query Video File List Data Format

Offset	Field	Data Type	Description
0	Logical Channel Number	BYTE	See Appendix B; 0 means all channels
1	Start Time	BCD[6]	YY-MM-DD-HH-MM-SS; all 0 means no start time condition
7	End Time	BCD[6]	YY-MM-DD-HH-MM-SS; all 0 means no end time condition
13	Alarm Flag	64BITS	bit0 ~ 31 see JT/T 808 alarm flag bit definition bit32 ~ 63 video alarm flag bit definition All 0 means no alarm type condition
21	Audio/Video Resource Type	BYTE	0: Audio/Video 1: Audio 2: Video 3: Video or Audio/Video

22	Stream Type	BYTE	0: All streams 1: Main stream 2: Sub-stream
23	Storage Type	BYTE	0: All storage 1: Main storage 2: Disaster backup storage

Example: 7E **92 05** 00 18 12 34 56 78 90 12 00 01 02 20 05 15 08 00 00 20 05 15 08 59 59 00 00 00 00 00 00 00 00 07 00 00 01 7E

0x1205 Terminal Uploads Audio/Video Resource List (Uplink)

The terminal responds to the platform's query audio/video resource list instruction using the terminal upload audio/video resource list message response. If the list is too large and needs to be transmitted in multiple packets, the packet splitting mechanism defined in JT/T 808 shall be used.

Table 22 Terminal Uploads Audio/Video Resource List Data Format

Offset	Field	Data Type	Description
0	Sequence Number	WORD	Sequence number corresponding to the query audio/video resource list instruction
2	Total Audio/Video Resources	DWORD	Set to 0 if there are no matching audio/video resources
6	Audio/Video Resource List		See Table 23

Table 23 Terminal Uploads Audio/Video Resource List Format

Offset	Field	Data Type	Description
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0	Logical Channel Number	BYTE	See Appendix B
1	Start Time	BCD[6]	YY-MM-DD-HH-MM-SS
7	End Time	BCD[6]	YY-MM-DD-HH-MM-SS
13	Alarm Flag	64BITS	bit0 ~ 31 according to JT/T 808 alarm flag bit definition bit32 ~ 63 video alarm flag bit definition
21	Audio/Video Resource Type	BYTE	0: Audio/Video 1: Audio 2: Video
22	Stream Type	BYTE	1: Main stream 2: Sub-stream
23	Storage Type	BYTE	1: Main storage 2: Disaster backup storage
24	File Size	DWORD	Unit: bytes (BYTE)

Example: 7E 12 05 00 22 12 34 56 78 90 12 00 01 00 01 00 00 00 01 01 25 05 06 10 00 00 25 05 06 12 00 00 00 00 00 00 00 00 00 01 01 01 00 00 00 00 BC 7E

0x9201 Platform Sends Remote Video Playback Request (Downlink)

The platform requests audio/video recording playback from the terminal device. After receiving the request, the terminal shall respond with 0x1205 (Terminal Uploads Recording File List) instruction, and then transmit recording data using the packet format defined in "RTP Protocol Payload Package Format for Real-time Audio/Video Stream Data Transmission".

Table 24 Platform Sends Remote Video Playback Request Data Format

Offset	Field	Data Type	Description
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0	Server IP Address Length	BYTE	Length n
1	Server IP Address	STRING	Real-time audio/video server IP address
1+n	Server Audio/Video Channel Listening Port (TCP)	WORD	Real-time audio/video server port number; set to 0 if TCP transmission is not used
3+n	Server Audio/Video Channel Listening Port (UDP)	WORD	Real-time audio/video server port number; set to 0 if UDP transmission is not used
5+n	Logical Channel Number	BYTE	See Appendix B
6+n	Audio/Video Type	BYTE	0: Audio/Video 1: Audio 2: Video 3: Video or Audio/Video
7+n	Stream Type	BYTE	0: Main stream or sub-stream 1: Main stream 2: Sub-stream If this channel only transmits audio, set this field to 0

8+n	Storage Type	BYTE	0: Main storage or disaster backup storage 1: Main storage 2: Disaster backup storage
9+n	Playback Mode	BYTE	0: Normal playback 1: Fast forward playback 2: Key frame fast reverse playback 3: Key frame playback 4: Single frame upload
10+n	Fast Forward or Fast Reverse Multiplier	BYTE	Valid when playback mode is 1 or 2; otherwise set to 0 0: Invalid 1: 1x 2: 2x 3: 4x 4: 8x 5: 16x
11+n	Start Time	BCD[6]	YY-MM-DD-HH-MM-SS; when playback mode is 4, this field represents the single frame upload time

17+n	End Time	BCD[6]	YY-MM-DD-HH-MM-SS; 0 means continuous playback; when playback mode is 4, this field is invalid
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Example: 7E **92 01** 00 24 12 34 56 78 90 12 00 01 0D 31 39 32 2E 31 36 38 2E 31 2E 31 30 30 1F 40 1F 41 01 00 00 00 00 00 25 05 06 10 00 00 25 05 06 12 00 00 18 7E

0x9202 Platform Sends Remote Video Playback Control (Downlink)

During audio/video recording playback, the platform can send playback control instructions to control the playback process.

Table 25 Platform Sends Remote Video Playback Control Data Format

Offset	Field	Data Type	Description
0	Audio/Video Channel Number	BYTE	See Appendix B
1	Playback Control	BYTE	0: Start playback 1: Pause playback 2: End playback 3: Fast forward playback 4: Key frame fast reverse playback 5: Drag playback 6: Key frame playback

2	Fast Forward or Fast Reverse Multiplier	BYTE	Valid when playback control is 3 or 4; otherwise set to 0 0: Invalid 1: 1x 2: 2x 3: 4x 4: 8x 5: 16x
3	Drag Playback Position	BCD[6]	YY-MM-DD-HH-MM-SS; valid when playback control is 5

Example: 7E 92 02 00 03 12 34 56 78 90 12 00 01 01 00 00 19 7E

0x9206 File Upload Instruction (Downlink)

The platform sends a file upload command to the terminal. After the terminal replies with a general response, it uploads the file to the specified path on the target FTP server via FTP.

Table 26 File Upload Instruction Data Format

Offset	Field	Data Type	Description
0	Server Address Length	BYTE	Length k
1	Server Address	STRING	FTP server address
1+k	Port	WORD	FTP server port number
3+k	Username Length	BYTE	Length l
4+k	Username	STRING	FTP username
4+k+l	Password Length	BYTE	Length m
5+k+l	Password	STRING	FTP password

5+k+l+m	File Upload Path Length	BYTE	Length n
6+k+l+m	File Upload Path	STRING	File upload path
6+k+l+m+n	Logical Channel Number	BYTE	See Appendix B
7+k+l+m+n	Start Time	BCD[6]	YY-MM-DD-HH-MM-SS
13+k+l+m+n	End Time	BCD[6]	YY-MM-DD-HH-MM-SS
19+k+l+m+n	Alarm Flag	64BITS	bit0 ~ 31 see JT/T 808 alarm flag bit definition bit32 ~ 63 video alarm flag bit definition All 0 means no specified alarm condition
27+k+l+m+n	Audio/Video Resource Type	BYTE	0: Audio/Video 1: Audio 2: Video 3: Video or Audio/Video
28+k+l+m+n	Stream Type	BYTE	0: Main stream or sub-stream 1: Main stream 2: Sub-stream
29+k+l+m+n	Storage Location	BYTE	0: Main storage or disaster backup storage 1: Main storage 2: Disaster backup storage

30+k+l+m+n	Task Execution Condition	BYTE	bit0: WIFI, set to 1 means download allowed under WIFI bit1: LAN, set to 1 means download allowed when LAN connected bit2: 3G/4G, set to 1 means download allowed under 3G/4G connection bit3: Whether to encrypt FTP username and password transmission, set to 1 means encrypt Encryption algorithm: Please contact software technical support team
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Example:7E 92 06 00 40 12 34 56 78 90 12 00 01 0D 31 39 32 2E 31 36 38 2E 31 2E 31 30 30 1F 40 04 74 65 73 74 04 31 32 33 34 0C 2F 75 70 6C 6F 61 64 2F 74 65 73 74 01 25 05 06 10 00 00 25 05 06 12 00 00 00 00 00 00 00 00 00 01 01 01 00 2F 7E

0x1206 File Upload Complete Notification (Uplink)

When all files have been uploaded via FTP, the terminal sends this instruction to notify the platform.

Table 27 File Upload Complete Notification Data Format

Offset	Field	Data Type	Description
0	Response Sequence Number	WORD	Sequence number corresponding to the platform file upload message

2	Result	BYTE	0: Success 1: Failure
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Example:7E **12 06** 00 03 12 34 56 78 90 12 00 01 00 01 00 9D 7E

0x9207 File Upload Control (Downlink)

The platform notifies the terminal to pause, resume, or cancel all files being transmitted.

Table 28 File Upload Control Data Format

Offset	Field	Data Type	Description
0	Response Sequence Number	WORD	Sequence number corresponding to the platform file upload message
2	Upload Control	BYTE	0: Pause 1: Resume 2: Cancel

Example:7E **92 07** 00 03 12 34 56 78 90 12 00 01 00 01 00 1C 7E

0x920F Query Recording Calendar Information (Downlink)

The platform queries the recording storage situation of the terminal device for a certain year and month.

Table 29 Query Recording Calendar Information Data Format

Offset	Field	Data Type	Description
0	Type	BYTE	0: All 1: Audio 2: Video
1	Year Month Calendar	BCD[2]	YY-MM

Example:7E **92 0F** 00 03 12 34 56 78 90 12 00 01 00 26 05 36 7E

0x120F Recording Calendar Information Reply (Uplink)

The terminal device replies the platform about the recording storage situation for a certain year and month.

Table 30 Recording Calendar Information Reply Data Format

Offset	Field	Data Type	Description
0	Calendar with Recording Files	32BITS (4 bytes)	bit0~31, bitwise; when value is 0, it means no recording file for that day; when value is 1, there is a recording file For example: bit7=1 means there is a recording file on the 8th of the queried month
4	Response Sequence Number	WORD	Response sequence number corresponding to the issued sequence number

Example:7E **12 0F** 00 06 12 34 56 78 90 12 00 01 20 83 10 00 01 00 22 7E

0x9215 Query Playback Resources and Event Point List (Downlink)

This instruction is sent to the device for query, returning both the playback resource list and the event point list (the front-end renders the resource list based on time points and event duration). Currently used for AOV (Always On Video) feature cameras.

Table 31 Query Playback Resources and Event Point List Data Format

Offset	Field	Data Type	Description
0	Logical Channel Number	BYTE	Query channel number; 0 means all channels

1	Start Time	BCD[6]	YY-MM-DD-HH-MM-SS
7	End Time	BCD[6]	YY-MM-DD-HH-MM-SS
13	Alarm Flag	64BITS	
21	Audio/Video Resource Type	BYTE	0: Audio/Video 1: Audio 2: Video 3: Video or Audio/Video
22	Stream Type	BYTE	0: All streams 1: Main stream 2: Sub-stream
23	Storage Type	BYTE	0: All storage 1: Main storage 2: Disaster backup storage

Example:7E 92 15 00 18 12 34 56 78 90 12 00 01 01 26 05 06 10 00 00 26 05 06 12 00 00 00 00 00 00 00 00 00 00 01 01 17 7E

0x1215 Terminal Uploads Audio/Video Resource List + Event Point List (Uplink)

The terminal responds to the platform's 0x9215, returning the audio/video resource list and marked event point list within the query range.

Table 32 Terminal Uploads Audio/Video Resource List + Event Point List Data Format

Offset	Field	Data Type	Description
0	Sequence Number	WORD	Sequence number corresponding to 0x9215 instruction

2	Total Audio/Video Resources	DWORD	Set to 0 if there are no matching audio/video resources
6	Audio/Video Resource List		See Table 33
x	Total Event Point List	DWORD	Set to 0 if there are no matching event point list
x+4	Event Point List		See Table 34

Table 33 Audio/Video Resource List

Offset	Field	Data Type	Description
0	Logical Channel Number	BYTE	
1	Start Time	BCD[6]	YY-MM-DD-HH-MM-SS
7	End Time	BCD[6]	YY-MM-DD-HH-MM-SS
13	Alarm Flag	64BITS	bit0~31 according to JT/T 808-2011 Table 18 alarm flag bit definition bit62: aov
21	Audio/Video Resource Type	BYTE	0: Audio/Video 1: Audio 2: Video
22	Stream Type	BYTE	1: Main stream 2: Sub-stream
23	Storage Type	BYTE	1: Main storage 2: Disaster backup storage

24	File Size	DWORD	Unit: bytes (BYTE)
28	Reserved Field	4BYTE	

Table 34 Event Point List

Offset	Field	Data Type	Description
0	Logical Channel Number	BYTE	
1	Event Type	64BITS	bit0: aov key focus
9	Event Time Point	BCD[6]	YY-MM-DD-HH-MM-SS
15	Event Duration	WORD	Unit: seconds, default 60
17	Reserved Field	4BYTE	

Example:7E 12 15 00 3F 12 34 56 78 90 12 00 01 01 00 01 00 00 00 01 26 05 06 10 00 00 26 05 06 12 00 00 00 00 00 00 00 00 00 00 00 00 00 01 01 00 00 10 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 00 00 00 00 00 26 05 06 10 30 00 3C 00 00 00 00 00 00 99 7E

5 Audio/Video Stream Data Transmission

5.1 RTP Payload Packet Format

Real-time audio/video stream data transmission references the RTP protocol, using UDP or TCP as the carrier. The payload packet format extends the IETF RFC 3550 RTP definition by adding fields such as message sequence number, SIM card number, and audio/video channel number.

Table 19 Audio/Video Stream and Passthrough Data Transmission Protocol Payload Packet Format Definition

Offset	Field	Data Type	Description
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0	Frame Header Identifier	DWORD	Fixed as 0x30 0x31 0x63 0x64
4	V	2 BITS	Fixed as 2
4	P	1 BIT	Fixed as 0
4	X	1 BIT	RTP header extension bit, fixed as 0
4	CC	4 BITS	Fixed as 1
5	M	1 BIT	Marker bit, determines whether it is the boundary of a complete data frame
5	PT	7 BITS	Payload type, see Table 19
6	Packet Sequence Number	WORD	Initial value is 0; sequence number increments by 1 for each RTP data packet sent
8	SIM Card Number	BCD[6]	Terminal device SIM card number
14	Logical Channel Number	BYTE	See Appendix B
15	Data Type	4 BITS	0000: Video I-frame 0001: Video P-frame 0010: Video B-frame 0011: Audio frame 0100: Passthrough data

15	Sub-package Processing Mark	4 BITS	0000: Atomic packet, cannot be split 0001: First packet in sub-package processing 0010: Last packet in sub-package processing 0011: Middle packet in sub-package processing
16	Timestamp	BYTE[8]	Identifies the relative time of the current frame of this RTP data packet, unit: milliseconds (ms). When data type is 0100, this field is not present
24	Last I Frame Interval	WORD	Time interval between this frame and the last key frame, unit: milliseconds (ms). This field is not present when data type is non-video frame
26	Last Frame Interval	WORD	Time interval between this frame and the previous frame, unit: milliseconds (ms). This field is not present when data type is non-video frame

28	Data Body Length	WORD	Length of subsequent data body, not including this field
30	Data Body	BYTE[n]	Audio/video data or passthrough data, length shall not exceed 950 bytes

5.2 Audio/Video Stream Request URL

After the government video surveillance platform sends a real-time preview or remote playback request instruction to the enterprise video monitoring platform and receives a successful response, it obtains the IP address and port number of the audio/video streaming server. The government video surveillance platform client directly sends a URL instruction to the enterprise audio/video streaming server to establish a link and obtain audio/video stream data.

URL Format:

http://[Server IP Address]:[Port]/[License Plate Number].[License Plate Color].[Logical Channel Number].[Audio/Video Flag].[Time-limited Token]

Table 35 Audio/Video Stream Request URL Instruction Data Item Definition

Category	Field	Description
Address Attribute Information	Server IP Address	Audio/video streaming server IP address
	Port Number	Audio/video streaming service port number
	License Plate Number	Shall use UTF-8 encoding and be uniformly converted to application/x-www-form-urlencoded MIME format
	License Plate Color	See Appendix C

	Logical Channel Number	See Appendix B; 0 means all channels
	Audio/Video Flag	0: Audio/Video 1: Audio 2: Video
Supplementary Information	Time-limited Token	Generated by enterprise platform server, length is 64 ASCII characters, should be updated every 24 hours
	Location Identifier	Any moment's satellite positioning time and longitude/latitude within the vehicle's 5 minutes, format: YYYYMMDD-HHMMSS-NXX.XXXXXX-EXXX.XXXXXX

Appendix B: Logical Channel Number Definition

Logical channel numbers are used to identify audio/video channels, defined as follows:

Logical Channel Number	Channel Type	Description
0	—	All channels (used for broadcast or query)
1~32	Video	Video channels
33~64	Audio	Audio channels
65~127	Custom	Extended channels

Description:

- Logical channel numbers are custom-allocated by device manufacturers
- In terminal audio/video parameter settings (0x8103 message), the mapping between physical channels and logical channels is configured through the channel list

- Instructions such as real-time video requests specify target channels through logical channel numbers

Appendix C: License Plate Color Definition

License plate color is used to identify vehicle license plate color, defined as follows:

Value	Color	Description
0	Not Registered	No license plate hung
1	Blue	Small passenger vehicle license plate
2	Yellow	Large passenger vehicle license plate
3	Black	Consulate, consulate vehicle license plate
4	White	Police vehicle license plate
5	Green	New energy vehicle license plate
6	Yellow (Double-layer)	Double-layer large passenger vehicle license plate
9	Other	Other types of license plates

Appendix A: Message Reference Table

Table A.1 Message Reference Table between Video Terminal and Video Platform

No.	Message Body Name	Message ID	No.	Message Body Name	Message ID
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1	Query Terminal Audio/Video Attributes	0x9003	8	File Upload Instruction	0x9206
2	Terminal Uploads Audio/Video Attributes	0x1003	9	File Upload Complete Notification	0x1206
3	Real-time Audio/Video Transmissio n Request	0x9101	10	File Upload Control	0x9207
4	Audio/Video Real-time Transmissio n Control	0x9102	11	Query Recording Calendar Information	0x920F
5	Real-time Audio/Video Stream and Passthrough Data Transmissio n	—	12	Recording Calendar Information Reply	0x120F
6	Query Resource List	0x9205	13	Query Playback Resources and Event Point List	0x9215
7	Terminal Uploads Audio/Video Resource List	0x1205	14	Terminal Uploads Audio/Video Resource List + Event Point List	0x1215